

4	(a)	$(5)(20) \neq (100)(1.2)$ R IS not inversely proportional to light intensity. Accept % difference, with justification	M1 A0
	(b)	When the potential difference across the LDR is 7.0 V, potential difference across the 1.2 kΩ resistor = 2.0 V Using $V = IR$ $2.0 = (1.2 \times 10^3) I$ $I = 1.67 \times 10^{-3} \text{ A}$ $R = \frac{V}{I} = \frac{7.0}{1.67 \times 10^{-3}} = 4200 \text{ } \Omega \text{ or } 4.2 \text{ k}\Omega$ Alternatively, use potential divider method to determine R.	C1 C1 A1
	(c)	From Fig. 4.1 , light intensity = 24 W m^{-2}	A1
	(d)	Length of strip, $\ell = (10 \times 5.0 \times 10^{-3}) + (10.0 \times 10^{-3}) = 0.060 \text{ m}$ Using $R = \frac{\rho \ell}{A}$, $\rho = \frac{RA}{\ell} = \frac{4200 \times 5.0 \times 10^{-7}}{0.060} = 3.5 \times 10^{-2} \text{ } \Omega \text{ m}$	C1 A1

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